MICROBIOLOGY

The evolution of antibiotic resistance

Clinically relevant evolution studies are needed to help fight the spread of antibiotic resistance

By **R. Craig MacLean**[1] *and* **Alvaro San Millan**[2]

For most of human history, bacterial pathogens have been a major cause of disease and mortality. The development of antibiotics provided a simple and effective treatment for bacterial infections, and antibiotics have since had huge effects on human health and longevity. These are threatened with the rise of antibiotic resistance (ABR): Many pathogenic bacteria have evolved resistance to the main classes of antibiotics, and multidrug-resistant bacteria have caused untreatable infections. ABR already imposes substantial health and economic burdens, and the global annual cost of ABR could increase to 10 million deaths and US$100 trillion by 2050 (*1*). Understanding how ABR evolves and spreads is therefore key to improving antibiotic treatment strategies.

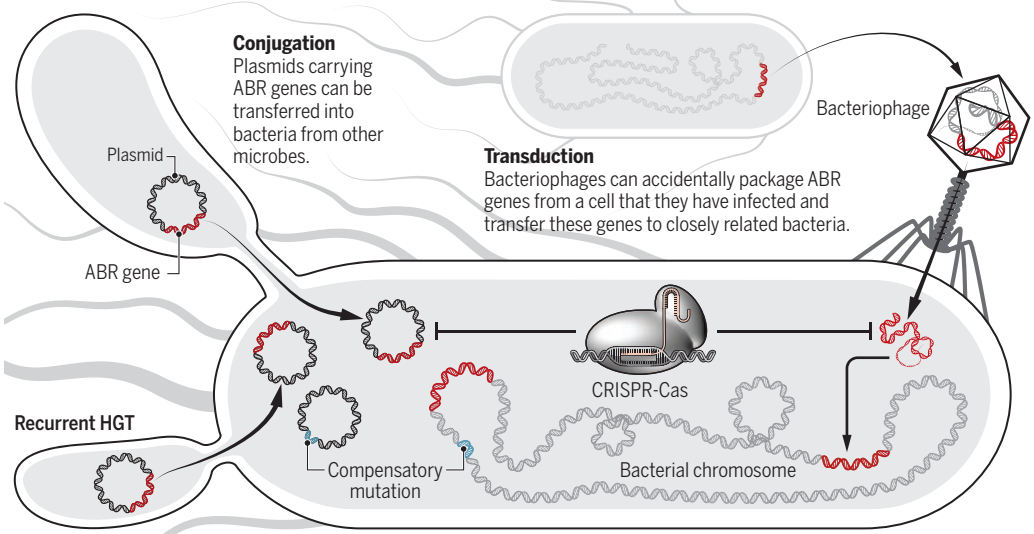
Two main approaches have been used to understand the processes driving the spread of ABR. Clinical studies have focused on DNA sequencing to identify the genes underlying ABR and the epidemiological dynamics of resistance. Experimental studies have focused on simple in vitro model systems to investigate how ABR changes in response to defined and controlled selective pressures, often supported by mathematical models (*2*). However, it has become increasingly clear that there are substantial gaps between these two approaches. Experimental studies have not placed enough emphasis on understanding the evolution of successful strains that have driven the rise of resistance in the clinic. To bridge these gaps and identify strategies to counteract ABR, experimental studies should shift focus toward investigating clinically relevant resistance genes and work with more realistic infection models.

Pathogenic bacteria acquire ABR through two fundamentally different genetic mechanisms. Spontaneous mutations that occur during the replication of the bacterial chromosome can make bacteria resistant to antibiotics, usually by modifying the cellular targets of antibiotics. Most pathogenic bacteria live in complex, densely populated microbial communities that are associated with human tissues, such as the gut, skin, and respiratory tract. The overwhelming majority of bacteria are harmless commensals that do not cause disease. However, it has become clear that these communities of commensal bacteria provide a rich source of dedicated ABR genes (*3*). Mobile genetic elements, such as plasmids and bacteriophages (bacterial viruses), transfer these ABR genes between bacterial populations through the process of horizontal gene transfer (HGT) (see the figure). Crucially, HGT allows pathogenic bacteria to evolve resistance by acquiring preexisting ABR genes from commensal bacteria. The basic mechanisms of HGT were discovered more than 50 years ago, and the current challenges are to better understand the rate at which pathogens acquire ABR genes by HGT and to identify the key conduits that transfer ABR genes from commensal bacteria to pathogens. For example, recent studies have identified previously unknown pathways for HGT, such as lateral transduction, which drives the rapid transfer of chromosomal DNA by bacteriophages (*4*). It is possible that studying patterns of distribution of ABR genes in microbiomes combined with experiments that track the acquisition of ABR genes in complex communities could help to start solving this puzzle.

One of the key insights from clinical microbial sequencing studies is that the rise of ABR in most of the important human pathogens has been largely driven by the spread of a small number of successful strains that can be effectively transmitted between patients. The acquisition of mobile elements that confer resistance to multiple antibiotics appears to be the key genetic mechanism that has enabled the dissemination of these globally prevalent ABR superbugs (*5*). For example, *Staphylococcus aureus* has evolved by acquiring staphylococcal cassette chromosome *mec* (SCC*mec*) elements that confer resistance to a wide variety of antibiotics, most notably methicil-

Mechanisms of mobile antibiotic resistance

Pathogenic bacteria can acquire antibiotic resistance (ABR) genes through two main mechanisms of horizontal gene transfer (HGT): conjugation and transduction. Compensatory mutations alleviate the fitness costs imposed by ABR genes, contributing to their stabilization. CRISPR-Cas–based systems may selectively destroy DNA carrying mobile ABR genes.



GRAPHIC: A. KITTERMAN/SCIENCE

[1]Department of Zoology, University of Oxford, Oxford, UK. [2]Department of Microbiology, Hospital Universitario Ramon y Cajal, Instituto Ramon y Cajal de Investigacion Sanitaria (IRYCIS) and Centro de Investigacion Biomedica en Red de Epidemiologia y Salud Publica (CIBERESP), Madrid, Spain. Email: craig.maclean@zoo.ox.ac.uk

Downloaded from https://www.science.org at Cambridge University on March 05, 2026

Published by AAAS

lin. Additionally, the dominant enteric pathogens *Escherichia coli* and *Klebsiella pneumoniae* have evolved resistance to the most recent generation of β-lactam antibiotics by acquiring plasmids carrying extended-spectrum β-lactamases and carbapenemases (*6*). Experimental studies have largely focused on studying the evolution of ABR by chromosomal mutations, often focusing on resistance mutations that have little, if any, clinical relevance. ABR is entirely caused by mutation in *Mycobacterium tuberculosis*, and mutations that confer resistance to fluoroquinolone antibiotics are important in the emergence of a broad spectrum of multidrug-resistant pathogens. However, it is important to note the discrepancy between these disciplines, with clinical studies focused on mobile elements and experimental studies focusing on mutational ABR.

Some strains of the enteric bacterium *Escherichia coli* (orange) can cause harmful infections and have acquired antibiotic resistance through plasmids.

PHOTO: GARY GAUGLER/SCIENCE SOURCE

A key insight from experimental studies has been that the acquisition of ABR, either by mutation or by HGT, creates a fitness cost that is reflected in reduced growth rate and virulence (*7*, *8*). This cost is thought to be the main obstacle that limits the spread of ABR by restricting the ability of resistant bacteria to successfully transmit to new hosts and establish infections. Pioneering experimental studies demonstrated that bacteria and/or mobile elements can evolve compensatory adaptations that eliminate the cost of resistance (*7*). Compensatory evolution allows ABR to stably persist in bacterial populations, even in the absence of antibiotic use, suggesting that compensation may be a key mechanism that drives the spread of successful ABR strains (*9*). A widely advocated solution to the ABR crisis is to restrict consumption, but the efficacy of this intervention is likely to be strongly dependent on the fitness of ABR bacteria. For example, reducing antibiotic use could have minimal effects on the prevalence of ABR if compensatory evolution has already occurred in dominant resistant strains.

Hundreds of studies have investigated the fitness of laboratory-evolved ABR, but understanding of the fitness of clinical pathogens remains limited (*8*). For example, few studies have measured the fitness of "naturally" evolved ABR in clinically isolated strains, and evidence of compensatory evolution has only been found in a few pathogens, such as *M. tuberculosis* (*10*). This is particularly important for plasmid-mediated ABR: Even though plasmids are the most important mechanism of ABR acquisition in pathogens (*5*), the fitness effects of important ABR plasmids in their natural bacterial hosts is unknown. Clinical strains are difficult to manipulate genetically, making it challenging to measure the contribution of individual ABR genes to fitness using existing methods. Technological developments, such as CRISPR-based genomic engineering, allow gene editing and even the complete removal of ABR plasmids from clinical strains, potentially resolving this limitation (*11*). A further challenge will be to move beyond the broth culture systems that are currently used to experimentally measure fitness by developing model systems that capture key elements of the infection biology of bacterial pathogens. Ideally, these systems should include abiotic stressors that pathogens encounter during infections (for example, limited resources) as well as interactions with commensal microbes that may promote or restrict the growth of ABR pathogens.

One widely advocated strategy for stopping the spread of ABR is to use multidrug combination therapies to suppress or reverse selection for resistance (*12*). Recent work has focused on understanding how collateral sensitivity, which occurs when increased resistance to one antibiotic confers increased sensitivity to a second antibiotic, can be used to design antibiotic combinations that select against resistance. Important mobile ABR genes confer collateral sensitivities (*13*), and ABR plasmids produce common metabolic alterations in their bacterial hosts (*14*), suggesting that it might be possible to identify exploitable collateral sensitivity to clinically important mobile genetic elements.

An alternative approach to combat HGT-acquired ABR is to exploit naturally occurring barriers to the transfer of ABR genes. Bacteria have an arsenal of xenogenic defense mechanisms, including both innate (restriction-modification) and adaptive (CRISPR-Cas) immune systems that recognize and destroy incoming "nonself" DNA (*15*). The native function of these systems is to protect bacteria from parasitic mobile elements, such as lytic bacteriophages, but they can be engineered to generate sequence-specific antimicrobials that target mobile ABR elements (*11*). In particular, CRISPR-Cas–based systems have the potential to genetically manipulate pathogen populations. A key challenge of using this approach is to develop tools to effectively deliver engineered CRISPR-Cas systems to pathogenic bacteria. ABR illustrates the incredible power of mobile genetic elements to introduce new genes into pathogen populations, and an improved understanding of HGT of resistance could ultimately help pave the way for the efficient delivery of "antiresistance" systems. Understanding the evolutionary drivers of ABR could make a substantial contribution to preserving the efficacy of next-generation antimicrobials, but realizing this potential will require a fundamental shift in how evolutionary biologists think about, and study, ABR. ■

REFERENCES AND NOTES

1. J. O'Neill, "Tackling drug-resistant infections globally: Final report and recommendations," *The Review on Antimicrobial Resistance* (2016).
2. M. Baym *et al.*, *Science* **353**, 1147 (2016).
3. M. O. A. Sommer, G. Dantas, G. M. Church, *Science* **325**, 1128 (2009).
4. J. Chen *et al.*, *Science* **362**, 207 (2018).
5. S. Baker, N. Thomson, F.-X. Weill, K. E. Holt, *Science* **360**, 733 (2018).
6. S. R. Partridge, S. M. Kwong, N. Firth, S. O. Jensen, *Clin. Microbiol. Rev.* **31**, e00088-17 (2018).
7. D. I. Andersson, D. Hughes, *Nat. Rev. Microbiol.* **8**, 260 (2010).
8. T. Vogwill, R. C. MacLean, *Evol. Appl.* **8**, 284 (2015).
9. A. San Millan *et al.*, *Nat. Commun.* **5**, 5208 (2014).
10. I. Comas *et al.*, *Nat. Genet.* **44**, 106 (2011).
11. D. Bikard *et al.*, *Nat. Biotechnol.* **32**, 1146 (2014).
12. M. Baym, L. K. Stone, R. Kishony, *Science* **351**, aad3292 (2016).
13. C. E. H. Rosenkilde *et al.*, *Nat. Commun.* **10**, 618 (2019).
14. A. San Millan *et al.*, *ISME J.* **12**, 3014 (2018).
15. S. Doron *et al.*, *Science* **359**, eaar4120 (2018).

ACKNOWLEDGMENTS

R.C.M. is supported by the Wellcome Trust (WT106918AIA), and A.S.M. is supported by the European Research Council (ERC grant 757440-PLASREVOLUTION).

10.1126/science.aax3879

Downloaded from https://www.science.org at Cambridge University on March 05, 2026

Published by AAAS

The evolution of antibiotic resistance

R. Craig MacLean and Alvaro San Millan

Science **365** (6458), . DOI: 10.1126/science.aax3879

View the article online
https://www.science.org/doi/10.1126/science.aax3879
Permissions
https://www.science.org/help/reprints-and-permissions

Downloaded from https://www.science.org at Cambridge University on March 05, 2026

Use of this article is subject to the Terms of service

Science (ISSN 1095-9203) is published by the American Association for the Advancement of Science. 1200 New York Avenue NW, Washington, DC 20005. The title *Science* is a registered trademark of AAAS.

Copyright © 2019 The Authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original U.S. Government Works